

JEE MAIN REVISION MASTER: LIMITS, CONTINUITY AND DIFFERENTIABILITY

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 29 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Evaluation of Limits & L'Hopital's Rule	16 PYQs	Limits
Very High	Continuity & Differentiability at a Point	15 PYQs	Continuity & Differentiability

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Evaluation of Limits & L'Hopital's Rule	$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$, $\lim_{x \rightarrow a} f(x)^{g(x)} = e^{\lim_{x \rightarrow a} (f(x) - 1)g(x)}$ (for 1^∞)
Continuity & Differentiability at a Point	$\text{Continuous at } a \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$, $\text{Differentiability} \implies \text{Continuity}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Evaluation of Limits & L'Hopital's Rule (Very High Yield — 16 PYQs)

Core Principle: Resolution of indeterminate forms ($0/0$, ∞/∞ , 1^∞), standard trigonometric and exponential limits, and Series expansion.

■ **Caution / JEE Pitfall:** Before applying L'Hopital's rule, ALWAYS check that the limit is in an indeterminate form ($0/0$ or ∞/∞). Differentiating a determinate form produces an incorrect answer.

■ **Speed Trick:** Maclaurin expansions are often faster than repeated L'Hopital: $\sin x = x - (x^3/6)$, $\cos x = 1 - (x^2/2)$, $e^x = 1 + x + (x^2/2)$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "If $\lim_{x \rightarrow 0} \frac{x^2 - ax - 3bx - 3}{x} = 5$, then $a+b=...$ " [Ans: C]
- [2026 Shift] Q#3 (MCQ): "If $f(x) = \begin{cases} ax+7 & \text{if } x < 1 \\ 3x-1 & \text{if } x = 1 \\ b/x+3 & \text{if } x > 1 \end{cases}$ is continuous at $x=...$ " [Ans: D]
- [2025 Shift] Q#4 (MCQ): " $\lim_{x \rightarrow 1} \frac{x^4 - \sqrt{x}}{\sqrt{x} - 1}$ is..." [Ans: N/A]

3.2 Continuity & Differentiability at a Point (Very High Yield — 15 PYQs)

Core Principle: Epsilon-delta and limit criteria for continuity, left/right hand derivatives, and graphical identification of non-differentiable sharp points.

■ **Caution / JEE Pitfall:** Differentiability implies continuity, but continuity DOES NOT imply differentiability (e.g. $f(x) = |x|$ is continuous at $x=0$ but not differentiable).

■ **Speed Trick:** The greatest integer function $[x]$ is discontinuous at all integer points $x \in \mathbb{Z}$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "If $f(x) = x^2 - 1$ if $x \geq 2$ and $x + 1$ if $x < 2$, then $\lim_{x \rightarrow 2} f(x) =$..." [Ans: B]
- **[2026 Shift] Q#3 (MCQ):** "If $f(x) = \begin{cases} ax + 7 & \text{if } x < 1 \\ 3x - 1 & \text{if } x = 1 \\ b / (x + 3) & \text{if } x > 1 \end{cases}$ is continuous at $x = 1$." [Ans: D]
- **[2025 Shift] Q#1 (MCQ):** "Match the following: In the following, $[x]$ denotes the greatest integer less than or equal to x .
Column - I Column - II (a) $x | x | x | x \dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If $f(x) = x^2 - 1$ if $x \geq 2$ and $x + 1$ if $x < 2$, then $\lim_{x \rightarrow 1} f(x) + \lim_{x \rightarrow 2} f(x) =$

- (A) 3
- (B) 5 [Correct]
- (C) 7
- (D) 9

Correct Answer: B

Solution Analysis: For $x \rightarrow 1$, the values of x are near 1, which is less than 2. So we use $f(x) = x + 1$. Hence, $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2$. Now for $x \rightarrow 2$, we check both sides. For $x < 2$, $f(x) = x + 1$. So, $\lim_{x \rightarrow 2^-} f(x) = 2 + 1 = 3$. For $x \geq 2$, $f(x) = x^2 - 1$. So, $\lim_{x \rightarrow 2^+} f(x) = 2^2 - 1 = 4 - 1 = 3$. Since both left-hand and right-hand limits are equal, $\lim_{x \rightarrow 2} f(x) = 3$. Now, $\lim_{x \rightarrow 1} f(x) + \lim_{x \rightarrow 2} f(x) = 2 + 3 = 5$. So the correct

Benchmark Problem 2: KCET 2026 (Q#2)

If $\lim_{x \rightarrow 3} \left(\frac{x^2 - ax - 3bx - 3}{x - 3} \right) = 5$, then $a + b =$

- (A) 1
- (B) 2
- (C) 3 [Correct]
- (D) 4

Correct Answer: C

Solution Analysis: For the limit $\lim_{x \rightarrow 3} \left(\frac{x^2 - ax - 3bx - 3}{x - 3} \right) = 5$ the denominator becomes 0 when $x = 3$. For the limit to be finite, the numerator must also become 0 at $x = 3$. So, $3^2 - a(3) - 3b(3) - 3 = 0$ $9 - 3a - 9b - 3 = 0$ Divide by 3 : $3 - a - 3b = 0$ $a + 3b = 3$ So the required value is 3 Correct option: C.